

Bowen Yuan (Bourne)

Location: Adelaide, Australia

Phone: 0452 516 601

Email: BowenYuan13@gmail.com

LinkedIn: <https://www.linkedin.com/in/bowen-yuan2020/>

Website: <https://empathiccomputing.org/team/bowen-yuan/>

ORCID: 0000-0002-3012-4077

Academic Profile

Computational design and extended reality researcher specialising in immersive technologies, digital communication methods, and spatial computing systems for architectural and urban contexts. My research develops spatio-temporal authoring frameworks and adaptive mixed reality environments that enable dynamic engagement with spatial narratives, environmental data, and procedural knowledge. My work bridges Human-Computer Interaction (HCI), Extended Reality (XR), computational design, and AI-augmented systems.

Education

PhD in Computer Science

University of South Australia (Adelaide University), Apr 2022 – Expected May 2026

Thesis: Dependency-Aware Mixed Reality Guidance

Supervisors: Assoc. Prof. Gun Lee; Prof. Mark Billinghurst

Visiting Researcher (Diminished Reality)

City University of Hong Kong, May 2025 – Sep 2025

Master of Information Technology

University of Queensland, Feb 2018 – Dec 2019

Bachelor of Electrical and Electronic Engineering

University of Liverpool, UK, Sep 2015 – Jul 2017

Academic Appointments

Lecturer (Sessional)

University of South Australia | Adelaide & Xi'an, Feb 2024 – Jul 2024

- Delivered studio-based Design Thinking curriculum.
- Led transnational teaching delivery in Xi'an.
- Designed workshops focused on user-centred design and spatial interaction systems.
- Provided curriculum alignment and assessment moderation.

Sessional Academic (Tutor & Practical Supervisor)

University of South Australia, Apr 2022 – Present

Courses: Data Structures Essentials; Cloud and Concurrent Programming; Introduction to Programming (Python).

- Delivered tutorials and laboratory sessions (20+ students).
- Explained advanced algorithmic concepts using visual modelling and live coding.
- Marked assessments and examinations with detailed technical feedback.

Research Contributions

- Developed STAGE framework for hierarchical dependency-graph AR replay systems.
- Designed immersive digital communication systems for architectural engagement.
- Engineered LLM-integrated Unity-based virtual agent systems.

Industry Experience

Software Developer

ByAsia Food Import Ltd., Sydney, Aug 2020 – Aug 2021

- Designed Warehouse Management System (WMS).

- Developed React Native logistics tracking application.
- Managed SQL databases and enterprise data integration.

Technical Skills

Programming: C++, Python, C#, Java, JavaScript (ES6), HTML/CSS

Frameworks: Unity3D, MRTK, React Native, React

AI & Systems: LLM API Integration, Python Torch

Academic Service & Engagement

- Peer reviewer for XR / HCI venues.
- International research collaboration (City University of Hong Kong).

Publication List

1. Yuan, B., Xiao, Z., Cho, H., Teo, T., Lee, G. A., Billinghamurst, M. "Dependency-Aware AR Guidance for Replaying Non-Linear Work, " In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems 2026 (Poster), Barcelona, Spain, 2026.
2. Yuan, B., Xiao, Z., Cho, H., Teo, T., Lee, G. A., Billinghamurst, M. "Location-Based Non-Linear Playback of AR Task Recording, " In XRMemory, held in conjunction with the IEEE Conference on Virtual Reality and 3D User Interfaces 2026, Daejeon, South Korea, 2026
3. Lee, G. A., Yuan, B., Hart, J. D., et al. "Facilitating Discussions in Immersive Virtual Meetings with Virtual Agents,". In Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces 2026 (Poster), Daejeon, South Korea, 2026.
4. B. Yuan, H. Cho, T. Teo, G. A. Lee and M. Billinghamurst, "Focus-Aware Task Guidance: Adaptive AR Instruction Playback Via Gaze and Location Tracking," *2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, Daejeon, Korea, Republic of, 2025, pp. 954-964, doi: <https://doi.org/10.1109/ISMAR67309.2025.00103>
5. Yuan, Bowen, et al. "Augmented reality spatial guidance cues for flexible multi-objective task environments." *Proceedings of the 19th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry*. 2024. DOI: <https://doi.org/10.1145/3703619.3706047>
6. Yuan, Bowen, et al. "The Fusion Nexus: Exploring the Confluence of Virtual and Real Worlds through Biocognitive Audio-Verbal Interface in Immersive XR Environments." *SIGGRAPH Asia 2023 XR*. 2023. 1-2. DOI: <https://doi.org/10.1145/3610549.3614594>
7. Arindam Dey, Amit Barde, Bowen Yuan, et al. "Effects of interacting with facial expressions and controllers in different virtual environments on presence, usability, affect, and neurophysiological signals." *International Journal of Human-Computer Studies* 160 (2022): 102762. DOI: <https://doi.org/10.1016/j.ijhcs.2021.102762>
8. H. Cho, Bowen Yuan, et al., "An Asynchronous Hybrid Cross Reality Collaborative System," *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, Sydney, Australia, 2023, pp. 70-73, doi: <https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00022>.
9. H. Cho, Bowen Yuan, et al., "Time Travellers: An Asynchronous Cross Reality Collaborative System," *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, Sydney, Australia, 2023, pp. 848-853, doi: <https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00186>.